

Benton Tripp

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Data scientist with more than six years of experience across local government, utilities, consulting, marketing, and forecasting. My work has included machine learning, statistical modeling, dashboards, data pipelines, GIS, and software development. I am comfortable working independently or with a team, explaining technical work clearly, and learning new tools and methods as projects require.

Work Experience

My experience includes solving technical and analytical problems with Python, R, SQL, GIS, and cloud-based tools. I have built and maintained machine learning models, data pipelines, dashboards, geospatial applications, and AI-assisted document analysis tools, including retrieval-augmented generation (RAG) workflows. I focus on producing clear, well-documented work that others can understand, maintain, and extend.

Technology Analyst

City of Raleigh

Raleigh, NC

June 2025 - Present

- Analyze department systems and processes, build dashboards, and automate recurring work for Planning and Development.
- Built a PyTorch computer vision application that detects and redacts engineering seals and other sensitive information in large PDF plan sets, reducing manual review.
- Produce city population estimates and long-range forecasts using census data, housing inventories, permit activity, and spatial modeling.
- Developed public and internal Power BI dashboards covering review timelines, permitting, inspections, annexations, workload, and development activity, including more than 20 internal dashboards used across multiple divisions.
- Analyzed development and demographic trends, including suburban and urban growth patterns presented to City Council and senior leadership.
- Built workflows for meeting tracking, event management, and customer intake, and prepared documentation and training for staff using department dashboards and data tools.

Data Scientist

Tennessee Valley Authority

Chattanooga, TN (Remote)

August 2022 - June 2025

- Developed machine learning models, statistical analyses, data pipelines, and R Shiny applications for environmental, geochemical, and operational projects.
- Built applications for groundwater evaluation, geochemical modeling, and billing anomaly detection.

- Developed document analysis tools using large language models, retrieval-augmented generation, embeddings, and LangChain.
- Built technical prototypes using Python, R, SQL, AWS, Azure DevOps, and Dataiku to test new analytical methods and applications.
- Participated in an enterprise MLOps platform assessment covering Domino Data Lab, DataRobot, Dataiku, SAS, and AWS SageMaker, with emphasis on model monitoring, reproducibility, workflow management, and deployment.
- Developed and taught advanced R programming courses for technical and non-technical employees.

Data Analyst

Elder Research Inc.

Raleigh, NC

August 2021 - August 2022

- Built a Databricks machine learning pipeline for ETL, model training, and deployment that identified supplier slippage drivers for Church & Dwight and supported more than \$1 million in savings.
- Designed an AWS workflow for Chick-fil-A using Redshift, Athena, and Apache Airflow to automate reconciliation between inventory and sales forecasting systems.
- Developed classification, forecasting, recommendation, and data integration solutions for commercial clients using Python, R, SQL, and cloud platforms.
- Gathered requirements, documented methods, maintained code, checked data quality, and presented findings to technical and nontechnical teams.

Data Science Intern

Merkle Inc.

New York City, NY (Remote)

June 2021 - August 2021

- Developed customer segmentation and propensity models for targeted marketing campaigns using Python, Databricks, and statistical modeling.
- Built data pipelines and an audience overlap analysis for the 2021 Jaguar and Land Rover campaigns, improving targeting efficiency by 15%.
- Prepared features, trained models, generated propensity scores, and supported campaign analysis and audience selection.

Forecasting Analyst

Genworth

Lynchburg, VA

January 2020 - March 2021

- Developed time-series models to forecast call volume and staffing needs using Python and PostgreSQL.
- Modeled trends, seasonality, day-of-week patterns, and other factors affecting incoming and outgoing call volume.
- Updated forecasts during the COVID-19 pandemic and communicated results for staffing, scheduling, and operational planning.

Education

Bachelor of Science in Data Analytics

- Western Governor's University
- Graduation: Fall, 2021

Graduate Certificate in Statistics

- North Carolina State University
- Completion: Spring, 2024

Master of Science in Geospatial Information Science and Technology

- North Carolina State University
- Graduation: Spring, 2025

Skills

Programming Languages

- Python
- R
- SAS
- SQL/NoSQL
- JavaScript/HTML/CSS

Software/Technologies

- Git
- Unix/Linux
- AWS (Athena, Redshift, EC2, Lambda)
- Dashboards (Power BI, Shiny, Dash)
- GIS Software (Esri Products, QGIS, GRASS GIS)
- MLOps Software (Dataiku, DataRobot, MLflow, Domino Data Lab)

Other Skills/Experience

- ETL and Data Integration
- Statistics & Applied Mathematics
- Machine Learning
- Deep Learning (PyTorch, TensorFlow, Keras)
- Natural Language Processing (NLP)
- LLMs (Retrieval-Augmented Generation, LangChain)
- Forecasting
- Geospatial Analytics & Spatial Statistics

Certifications & Achievements

- AWS Cloud Practitioner, July 2024 - *Amazon Web Services (AWS)*
- IT Information Library Foundations Certification (ITIL), June 2021 - *AXELOS Global Best Practice*

Independent Research/Projects

Species Distribution Modeling

- Conducted comparative analysis between machine learning and traditional approaches (Inhomogeneous Poisson Point Process, MaxEnt) for Species Distribution Modeling, implementing all analytical pipelines in R.

- Analyzed distribution patterns of 8 bird species across 4 US states using bioclimatic variables, developing fully reproducible workflows with robust version control.

Utilizing the Clay Foundation Model and Sentinel-2 Imagery for Urban Growth Monitoring

- Developed proof-of-concept study integrating Sentinel-2 multispectral satellite imagery with the Clay Foundation Model (open-source deep Earth Observation foundation model) to monitor urban growth patterns using Python and PyTorch.
- Engineered a scalable, data-efficient framework generating urban imperviousness estimates with higher temporal resolution than standard datasets like the National Land Cover Database.
- Processed Sentinel-2 imagery into structured geospatial datasets and leveraged Clay Foundation Model for feature extraction and embedding generation.
- Designed and evaluated both simple and complex neural network architectures in PyTorch, enhancing model performance through systematic hyperparameter optimization to achieve high accuracy in urban imperviousness prediction.
- Demonstrated methodology's potential for enhancing urban monitoring frequency and responsiveness, directly supporting sustainable urban planning and resource management initiatives.

Personal Attributes

With a strong work ethic and a results-driven mindset, I am committed to personal and professional growth. I thrive in dynamic environments that challenge me to push the boundaries of my knowledge and skill set. Highly organized and detail-oriented, I am passionate about continuously learning and applying new technologies and analytical approaches to solve complex problems and contribute to organizational success.